Assignment No. 3

Title	Create a Java program demonstrating single inheritance
Aim/Problem Statement	Create a Java program demonstrating single inheritance where a subclass extends a superclass and calls its methods.
CO Mapped	CO 3
Pre-requisite	1. Basic programming skills in Java 2. 64-bit Open Source Linux (e.g., Ubuntu) 3. Open-source Java Development Kit (JDK), Java compiler (javac), or IDE (e.g., Eclipse, IntelliJ, VS code)
LearningObjective	To learn and apply Single level Inheritance concepts to develop concurrent applications.

Theory:

What is Inheritance?

Inheritance is a fundamental concept in ObjectOriented Programming (OOP). It is the mechanism in Java by which one class is allowed to inherit the features(fields and methods) of another class. In Java, Inheritance means creating new classes based on existing ones. A class that inherits from another class can reuse the methods and fields of that class.

Example: In the following example, Animal is the base class and Dog, Cat and Cow are derived classes that extend the Animal class.

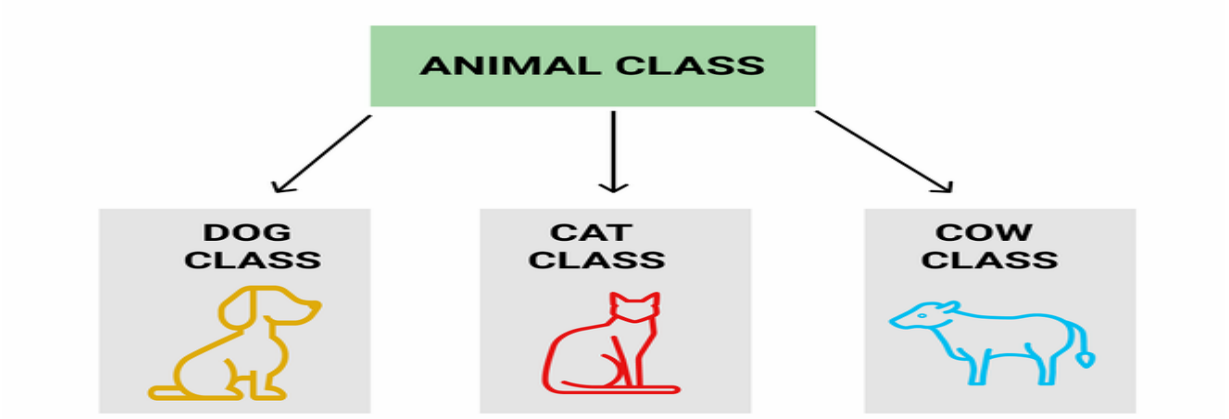

Figure: Base class derived to three subclasses

In Java, it is possible to inherit attributes and methods from one class to another. We group the "inheritance concept" into two categories:

- **subclass** (child) - the class that inherits from another class
- **superclass** (parent) - the class being inherited from
-
- **Subclass**
- A subclass, also known as **child class** or **derived class**, is the class that inherits the properties and behaviors of another class. So, if A and B are two classes and if B class inherits A class, then the B class is called the subclass **Superclass**
- A super class, also known as **parent class** or **base class**, is the class whose properties and behaviors are inherited by the subclass. So, if A and B are two classes and if B class inherits A class, then A class is called the superclass.

Types of Inheritance:

Below are the different types of inheritance which are supported by Java.

- Single Inheritance
- Multilevel Inheritance
- Hierarchical Inheritance
- Multiple Inheritance
- Hybrid Inheritance

Single Inheritance: In single inheritance, a sub-class is derived from only one super class. It inherits the properties and behavior of a single-parent class. Sometimes, it is also known as simple inheritance.

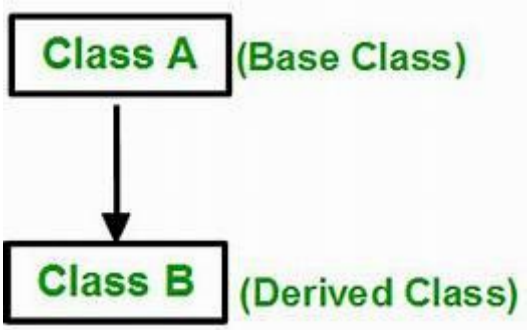

Figure: Single level inheritance (Base class A derived to single Subclass)

Syntax:

```
class Subclass-name extends Superclass-name
        {
        //methods and fields
        }
```

- The extends keyword indicates that you are making a new class that derives from an existing class. The meaning of "extends" is to increase the functionality.
- In the terminology of Java, a class which is inherited is called parent or super class and the new class is called child or subclass.

Algorithm:

1. Start the program.
2. Define a superclass (e.g., Person) with fields and methods.
3. Define a subclass (e.g., Student) that extends the superclass.
4. In the subclass, add additional fields and methods specific to Student.
5. In the main method, create an object of the subclass.
6. Call the methods of both superclass and subclass using the subclass object.
7. Display the results on the console.
8. Stop the program.

Conclusion:

Questions:

1. What is inheritance in Java?
2. Can a subclass inherit private members of a superclass? Why or why not?
3. Can we achieve single-level inheritance if the superclass constructor is private? Why or why not?